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Using Social Media to Identify the Effects of Congressional Viewpoints on Asset Prices

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1 Big picture

- **Question.** Do individual politicians affect stock prices when they publicly express viewpoints about specific firms?
- **Data source.** The paper uses official, campaign, and personal Twitter accounts of U.S. Senators and House members from January 2013 to December 2020.
- **Core identifying idea.** Tweets are timestamped to the second. The authors use high-frequency stock-price windows around tweets to isolate the market impact of politicians' viewpoints.
- **Sample.** Starting from 2.5 million congressional tweets from 740 Twitter accounts, the authors identify 11,602 tweets that explicitly convey opinions about 454 Russell 3000 firms.
- **Tone measure.** Tweets are classified as supportive or critical using the Loughran–McDonald positive and negative word dictionaries. The tone score is standardized.
- **Main high-frequency result.** A one-standard-deviation more supportive tone increases the mentioned firm's stock return by about 2 basis points within a six-minute window.
- **Persistence.** The effect becomes much larger when the window extends to one day: roughly 7–8 basis points depending on the return adjustment.
- **Portfolio evidence.** A daily-rebalanced long-short portfolio that buys firms receiving very supportive tweets and shorts firms receiving very critical tweets earns economically large monthly abnormal returns.
- **Mechanism.** The bulk of the effect comes from tweets that reveal news about future policy and legislation.
- **Committee meetings.** Price effects are amplified around congressional committee meetings, especially when the tweet comes from a committee member, resembles the meeting topic, or references a specific bill.
- **Industry spillovers.** Tweets tied to bills also move firms in the same industry, consistent with legislation affecting groups of firms rather than isolated companies.
- **Economic takeaway.** Congressional social media is not just political communication. It is a real-time channel through which policy information enters asset prices.

2 Data and measurement

2.1 Congressional Twitter data

- The authors collect tweets from official, campaign, and personal accounts listed on congressional websites.
- Accounts that cannot be linked to a single legislator, such as institutional accounts that change hands across terms, are excluded.
- The sample covers January 2013 through December 2020.
- The resulting data contain 2.5 million tweets, 740 accounts, and about 30 million words.
- The median member of Congress posts 42 tweets per month.

Interpretation. The Twitter setting is useful because posts have standardized text and precise timestamps. This permits event-study designs at minute-level frequency.

2.2 Identifying tweets about firms

- The firm universe is restricted to Russell 3000 companies, which cover roughly 98% of U.S. equity-market capitalization.

- A tweet is classified as firm-related if it includes the firm legal name or variants, the official company Twitter handle, the CEO’s name, or the CEO’s Twitter handle.
- The automated search generates 24,032 matches.
- The authors manually check all matches to eliminate false positives and cases where the company name appears without an opinion about the firm.
- The final sample contains 11,602 firm-targeting tweets about 454 unique firms.

Interpretation. Manual cleaning is central because company names often overlap with ordinary words or unrelated contexts. The final sample is designed to keep tweets that clearly express a politician’s viewpoint about a firm.

2.3 Tone measure

Let $P_{i,t}$ be the number of positive words and $N_{i,t}$ the number of negative words in tweet t about firm i . Let $W_{i,t}$ be the total word count. The raw relative tone measure is

$$Tone_{i,t}^{raw} = \frac{P_{i,t} - N_{i,t}}{W_{i,t}}. \quad (1)$$

The paper then standardizes this measure:

$$Tone_{i,t} = \frac{Tone_{i,t}^{raw} - \overline{Tone^{raw}}}{sd(Tone^{raw})}. \quad (2)$$

Interpretation. A higher value means a more supportive tweet; a lower value means a more critical tweet. Standardization makes coefficients interpretable as the response to a one-standard-deviation increase in tone.

2.4 High-frequency stock-price data

- High-frequency stock prices come from the NYSE Trade and Quote database.
- The benchmark event window is $[-1, +5]$ minutes around the tweet.
- The outcome is the log stock-price change from the last transaction before the tweet window to the first transaction after the tweet window.
- If no trades occur within the required before/after window, the observation is excluded as not informative for immediate price response.
- The average time gap between the before and after stock prices is 7 minutes and 25 seconds.

Interpretation. The high-frequency design aims to minimize the probability that other news systematically drives the stock price change exactly around the tweet.

2.5 Controls for other news

The paper controls for several types of news and market conditions.

- **Firm news:** earnings announcements, revenue announcements, management guidance, analyst forecast revisions, Bloomberg firm-news sentiment, news article headlines, and stock-related tweets from news accounts.
- **Macro news:** top Bloomberg macroeconomic indicators, including precise announcement times and surprise components.

- **Market conditions:** intraday cumulative return before the tweet, Amihud illiquidity, and intraday volatility.
- **Daily regressions:** controls are aggregated to daily frequency, with news after 4 p.m. assigned to the next day.

Interpretation. The controls address the concern that congressional tweets may coincide with ordinary corporate or macro news. The main coefficient remains similar after these controls.

2.6 Committee meeting and legislation data

- The paper uses information on House and Senate committee meetings and hearings.
- Meeting-day variables identify whether a firm-targeting tweet coincides with a committee meeting.
- Member variables identify whether the tweeting politician is a member of the committee holding the meeting.
- Text-similarity variables identify whether the tweet text resembles the committee meeting topic.
- Bill variables identify tweets linked to specific bills or joint resolutions discussed around committee meetings.
- The authors also use a Legislative Effectiveness Score to proxy for politically influential legislators.

Interpretation. Committee meetings are likely moments when future policy information is produced, debated, or revealed. If tweets reveal legislative news, effects should be stronger on these dates and for involved legislators.

3 Models and empirical specifications

3.1 Benchmark high-frequency regression

The main high-frequency regression is

$$\Delta p_{i,t} = a + b \cdot \text{Tone}_{i,t} + \varepsilon_{i,t}, \quad (1)$$

where $\Delta p_{i,t}$ is the log stock-price change in basis points for firm i around tweet time t .

Interpretation. The coefficient b measures the average stock-price response, in basis points, to a one-standard-deviation more supportive tweet. A positive b means supportive tweets increase stock prices and critical tweets reduce them.

3.2 Daily persistence regression

For daily returns, the paper estimates

$$r_{i,t} = a_0 + a_1 I_{i,t} + b \cdot (\text{Tone}_{i,t} \times I_{i,t}) + \varepsilon_{i,t}, \quad (2)$$

where $I_{i,t}$ equals one if a tweet about firm i occurs on day t .

Interpretation. The interaction isolates the effect of tone on tweet days. The dummy $I_{i,t}$ controls for the average return on days when any firm-targeting tweet occurs.

3.3 Return adjustments

The daily outcome is estimated using several return definitions:

$$Y_{i,t} \in \{r_{i,t}^{raw}, r_{i,t}^{CAPM}, r_{i,t}^{FF3}, r_{i,t}^{FF4}\}. \quad (3)$$

Interpretation. The core result is not driven by standard factor exposures. Tone effects remain after CAPM, Fama–French three-factor, and Fama–French–Carhart four-factor adjustments.

3.4 Portfolio strategy

The long-short portfolio strategy uses the recent distribution of tone.

- Buy stock i if the daily tone for firm i is above the 90th percentile of its prior 12-month tone distribution.
- Short stock i if the daily tone is below the 10th percentile.
- Form value-weighted long and short portfolios at the end of the day following the tweet.
- Rebalance daily and evaluate monthly returns and factor alphas.

Interpretation. The portfolio test asks whether the information in congressional tweets is economically relevant and persistent enough to generate abnormal returns after realistic timing.

3.5 Committee meeting interaction regressions

The meeting-day regressions extend the benchmark model as follows:

$$Y_{i,t} = \alpha_i + \beta_0 \text{Tone}_{i,t} + \beta_1 (\text{Tone}_{i,t} \times \text{Meeting}_t) + \Gamma' X_{i,t} + u_{i,t}, \quad (4)$$

and then add more refined interactions:

$$\text{Tone}_{i,t} \times \text{Meeting}_t \times \text{Member}_{i,t}, \quad (5)$$

$$\text{Tone}_{i,t} \times \text{Meeting}_t \times \text{Member}_{i,t} \times \text{TextSimilar}_{i,t}. \quad (6)$$

Interpretation. If committee meetings produce policy news, tone effects should be stronger when tweets occur on meeting days, when the tweeting politician sits on the relevant committee, and when the tweet text relates closely to meeting topics.

3.6 Bill and industry spillover regressions

The bill tests add interactions such as

$$\text{Tone}_{i,t} \times \text{Meeting}_t \times \text{Bill}_t \quad (7)$$

and

$$\text{Tone}_{i,t} \times \text{Meeting}_t \times \text{Bill}_t \times \text{LES}_{i,t}. \quad (8)$$

Interpretation. Tweets tied to specific bills should contain more precise policy information. If legislation affects industries rather than single firms, related firms in the same industry should also move.

4 Identification and empirical design

4.1 High-frequency identifying assumption

- The identifying assumption is that no other relevant information affecting the stock price is systematically released within the very narrow tweet window.
- The $[-1, +5]$ minute window is short enough to make systematic confounding less likely.
- Controls for firm-level and macro-level news further reduce the chance that ordinary news releases explain the result.
- The event-study figure shows no clear pre-trend in the minutes before the tweet.

Interpretation. The design is strongest for immediate price reactions. It is not based on random assignment of tweet tone, but the precise timing makes alternative explanations harder to sustain.

4.2 Why tone identifies directional information

- Critical and supportive tweets about the same type of target should move prices in opposite directions if they contain value-relevant information.
- The binned scatter and regression estimates show this monotonic relation.
- The result holds for raw and abnormal returns.

Interpretation. The sign of tone provides a direct prediction for the sign of the price movement. This makes the test sharper than simply asking whether tweets generate volatility.

4.3 Why persistence matters

- A six-minute window captures immediate reaction, but political news may take time to process.
- Daily regressions show larger effects than minute-level regressions.
- The event-study plot shows drift after the tweet and no subsequent reversal over the next several days.

Interpretation. Persistence supports the interpretation that tweets reveal information, rather than merely generating temporary attention or sentiment pressure.

4.4 Why committee meetings matter

- Committees are central to drafting, evaluating, and proposing legislation.
- Meetings and hearings are moments when specialized information is produced and circulated.
- Effects are stronger when tweets come from committee members and when tweet text is related to the meeting topic.

Interpretation. The committee evidence links the price response to future policy news, not just generic political attention.

4.5 What the paper cannot fully prove

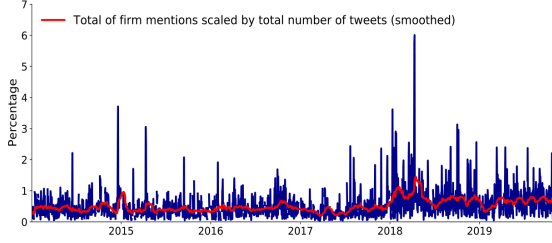
- Tweets are not randomly assigned across firms, politicians, or topics.
- Textual tone measures can misclassify sarcasm, context, or complex statements.
- The high-frequency design mitigates but does not eliminate the possibility of coincident information releases.
- Portfolio returns are informative about persistence, but they are not a trading-cost-adjusted implementation study.
- The paper identifies the effect of public viewpoints; it does not measure the private information politicians may have before tweeting.

Interpretation. The evidence is compelling because several independent patterns align: immediate price moves, daily persistence, long-short returns, meeting-day amplification, bill-related effects, and industry spillovers.

5 Tables and figures

The figures and tables below are directly cropped from the original paper and grouped to save space.

5.1 Figure 1 + Figure 2: Tweet frequency and tone-return relation



(a) Tweets by Congress members targeting firms

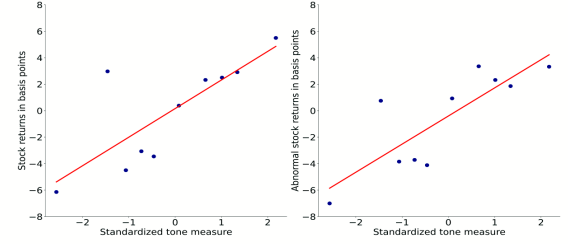


Figure 2

(b) Tone measure and stock returns

Read:

- Figure 1 shows that the frequency of firm-targeting congressional tweets varies strongly over time.
- The average firm-targeting share is around 0.5% of congressional tweets, roughly six firm-targeting tweets per day.
- Figure 2 shows a clear positive relation between standardized tone and high-frequency stock returns.
- The first tone decile has an average stock-price reduction of about -6.19 bps; the last decile has an average increase of about 4.84 bps.

Economic message. Congressional firm-specific tweets are common enough to study systematically, and their tone is strongly related to immediate stock-price responses.

5.2 Table 1 + Figure 3: High-frequency effects and event-study dynamics

Table 1
High-frequency stock prices responses to congressional viewpoints

Coefficient	Variable	Raw returns			Abnormal returns
		(1)	(2)	(3)	(4)
a	1	-0.07 [-0.19]	-0.46 [-1.36]	-0.46 [-1.37]	-0.25 [-0.40]
b	$Tone_{i,t}$	2.13 [2.94]	2.52 [2.87]	2.53 [2.89]	2.20 [2.81]
R -squared (%)		0.59	0.93	1.02	0.90
Observations		4,876	4,716	4,716	4,716
Firm-level controls	No	Yes	Yes	Yes	Yes
Macro-level controls	No	No	No	Yes	Yes

The table presents regression estimates for the equation: $\Delta p_{i,t} = a + b \cdot Tone_{i,t} + \epsilon_{i,t}$, where $\Delta p_{i,t}$ denotes the change in log stock prices for stock i around tweet t . A 6-minute window (1 minute before and 5 minutes after the event t) is used to calculate price changes. $Tone_{i,t}$ represents the tone measure. Stock-fixed effects are used in all regressions. Column 2 includes firm-level controls, column 3 adds macro-level controls to firm-level controls, and column 4 includes firm-level controls, macro-level controls, and abnormal returns, calculated as the difference between a company's stock return and the aggregate market return. Firm-level controls consist of intraday illiquidity measures, cumulative stock returns up to 10 minutes before the tweet, time-stamped stock news article headlines, stock-related tweets from news media accounts, analysts' cash flow revisions, and forecast errors. Macro-level controls consist of news announcements of the top-50 macroeconomic indicators. Event-based controls use a 20-minute window around the tweet (10 minutes before and 10 minutes after the stock prices used to compute $\Delta p_{i,t}$). Standard errors are clustered at the firm and day levels, with t -statistics in brackets. $Tone_{i,t}$ is standardized, while $\Delta p_{i,t}$ is in basis points. R -squared statistics are reported as percentages.

(a) High-frequency stock-price responses

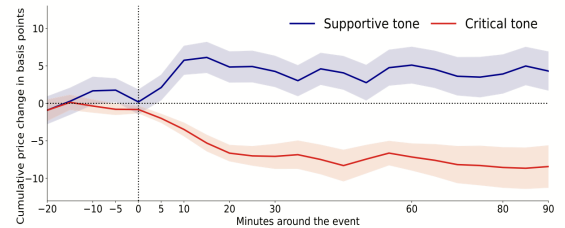


Figure 3

Event-study plot: Minutes around the event

This figure shows the effect of congressional tweets on stock prices across various event windows. Tweets are initially sorted into 10 deciles based on their tone measure. For each stock in a decile, cumulative returns are

(b) Minutes around the tweet

Read:

- Table 1 estimates that a one-standard-deviation higher tone raises raw high-frequency returns by 2.13–2.53 bps.

- With abnormal returns and both firm- and macro-level controls, the coefficient is 2.20 bps with a t -statistic of 2.81.
- The coefficient remains stable after controlling for news and market conditions.
- Figure 3 shows little pre-trend before the tweet.
- Supportive tweets generate positive cumulative returns, while critical tweets generate negative cumulative returns.
- The critical-tone effect reaches roughly -8.42 bps by the 90-minute window.

Economic message. The core high-frequency result is that politicians’ firm-specific viewpoints move prices within minutes, and the sign of the move follows the sign of the viewpoint.

5.3 Table 2 + Table 3: Daily persistence and portfolio returns

Table 2
Daily asset prices responses to congressional viewpoints

Coefficient	Variable	Raw returns (1)	Residualized returns using		
			CAPM (2)	FF3 (3)	FF4 (4)
a_0	1	-0.70 [-0.07]	-3.46 [-0.95]	-4.82 [-1.47]	-3.81 [-1.35]
a_1	$I_{i,t}$	-9.56 [-2.98]	-7.73 [-3.09]	-6.73 [-2.93]	-5.96 [-2.58]
b	$Tone_{i,t} \times I_{i,t}$	8.31 [2.94]	7.25 [3.01]	7.01 [2.89]	7.23 [2.91]
R -squared (%)		0.73	0.76	0.80	0.78
Observations		719,388	716,351	716,351	716,351
Firm-level controls	Yes	Yes	Yes	Yes	Yes
Macro-level controls	Yes	Yes	Yes	Yes	Yes

The table presents regression estimates for Equation (2): $r_{i,t} = a_0 + a_1 I_{i,t} + b \cdot (Tone_{i,t} \times I_{i,t}) + \epsilon_{i,t}$, where $r_{i,t}$ denotes the daily stock return for company i , $I_{i,t}$ is a dummy variable equal to one if a tweet about company i occurs on day t , and zero otherwise. $Tone_{i,t}$ represents the daily tone measure. All regressions use stock-fixed effects and incorporate firm-level and macro-level controls. Firm-level controls include daily illiquidity and volatility measures, weekly cumulative stock returns up to day $t-1$ before the tweet, Bloomberg news sentiment measure for company i on day t and $t-1$, daily tone measures for stock news article headlines and stock-related tweets from news media accounts, analysts’ cash flow revisions, and forecast errors. Macro-level controls consist of news announcements for the top-50 macroeconomic indicators. Column 1 reports results with using raw returns; column 2 with CAPM-adjusted returns; column 3 with Fama-French three-factor model-adjusted returns; and column 4 with Fama-French-Carhart four-factor model-adjusted returns. Stock returns are in basis points, and $Tone_{i,t}$ is standardized. The full panel of stock returns is used. Standard errors are clustered at the firm and day levels, with t -statistics in brackets. R -squared statistic is reported as percentage.

(a) Daily asset-price responses

Table 3
Performance evaluation: Mean returns and factor alphas

	Portfolio returns relative to the day of the tweet				
	Day $t+1$		Day $t-1$		Day t
	Short (1)	Long (2)	Long - Short (3)	Long - Short (4)	Long - Short (5)
Average return	-0.20	0.99	1.19	-0.02	1.30
Standard deviation	4.64	4.85	6.02	5.23	7.72
CAPM alpha	-0.84	0.40	1.13	0.10	1.83
	[-2.40]	[1.87]	[3.68]	[0.12]	[4.22]
FF3 alpha	-0.99	0.23	1.12	0.16	1.91
	[-1.98]	[1.48]	[3.32]	[0.20]	[4.95]
FF4 alpha	-1.04	0.10	1.03	0.18	2.11
	[-2.13]	[1.32]	[3.14]	[0.24]	[4.91]

This table presents monthly mean returns and monthly factor alphas for three distinct portfolios. The long (short) portfolio buys stock i if the corresponding tone measure is above (below) the 90th (10th) percentile of the tone distribution computed over the previous 12 months. We value-weight the stocks to create portfolios when multiple firms are assigned to the same leg on the same day. In columns 1 through 3, to execute the buy or sell order, we allow for a one-business-day gap between day t of the ranking period and day $t+1$ of the holding period. In columns 4 and 5, we execute the buy or sell order on day $t-1$ and on day t , respectively. The portfolios are rebalanced daily. Once we obtain the daily portfolio returns for the long and short sides, we accumulate the daily returns at the monthly frequency. The “long-short” portfolio buys the long portfolio and sells the short portfolio. The alphas represent the intercepts from time series regressions of the portfolio excess returns on factor alphas. The four factors include the aggregate market excess return, the size factor, the value factor, and the momentum factor. Standard errors adjust for heteroskedasticity and autocorrelation. Returns are in percent. We report the t -statistics in brackets.

(b) Long-short portfolio performance

Read:

- Table 2 shows that daily effects are larger than minute-level effects: the raw daily tone coefficient is 8.31 bps.
- Factor-adjusted daily effects remain large: 7.25 bps with CAPM adjustment, 7.01 bps with FF3 adjustment, and 7.23 bps with FF4 adjustment.
- Table 3 shows that the long-short portfolio earns 1.19% per month on average when formed on day $t+1$.
- The day $t+1$ long-short CAPM, FF3, and FF4 alphas are 1.13%, 1.12%, and 1.03% per month.
- Forming the portfolio on day t yields even larger alphas, while forming it on day $t-1$ yields no meaningful return.

Economic message. The information in congressional tweets is not a fleeting microstructure artifact. Prices continue to incorporate the information over the day and generate persistent abnormal returns.

5.4 Table 4 + Figure 4: Committee meetings and legislative information

Table 4
Stock price responses around committee meetings

Dependent variable	High-frequency abnormal returns				Daily FF4 adj. returns
	(1)	(2)	(3)	(4)	
<i>Tone_{i,t}</i>	2.20 [2.81]	1.70 [2.56]	1.70 [2.56]	1.70 [2.56]	6.05 [2.54]
<i>Tone_{i,t} × Meeting_t</i>		0.76 [2.65]	0.13 [0.50]	0.13 [0.50]	0.48 [1.01]
<i>Tone_{i,t} × Meeting and Member_{i,t}</i>			1.38 [2.49]	1.33 [2.21]	4.74 [1.87]
<i>Tone_{i,t} × Meeting, Member, and TextSimilar_{i,t}</i>				1.01 [2.67]	4.50 [2.31]
1	−0.25 [−0.40]	−1.04 [−1.31]	−1.04 [−1.30]	−1.04 [−1.30]	−6.76 [−1.52]
<i>Meeting_t</i>		1.28 [1.37]	0.90 [0.97]	0.90 [0.97]	3.08 [0.50]
<i>Meeting and Member_{i,t}</i>			0.98 [0.85]	1.62 [1.12]	−5.24 [−1.01]
<i>Meeting, Member, and TextSimilar_{i,t}</i>				−1.37 [−1.56]	−7.33 [−0.97]
R-squared (%)	0.90	0.92	0.91	0.92	0.83
Observations	4,716	4,716	4,716	4,716	11,355
Firm-level controls	Yes	Yes	Yes	Yes	Yes
Macro-level controls	Yes	Yes	Yes	Yes	Yes

Moving through columns 1 to 4, we progressively introduce more refined regressors. In column 2, *Meeting_t* is a dummy variable that equals one if a tweet coincides with a committee meeting and zero otherwise. In column 3, *Meeting and Member_{i,t}* is a dummy variable that equals one if a politician's tweet occurs on a day when his or her corresponding committee is holding a meeting and zero otherwise. In column 4, *Meeting, Member, and TextSimilar_{i,t}* is a dummy variable that equals one if the text similarity measure is above its median value and the politician belongs to the committee holding the meeting and zero otherwise. The dependent variable in columns 1 to 4 is abnormal returns, calculated as the difference between a company's stock return and the aggregate market return using a 6-minute window around the tweet. In column 5, the dependent variable is the Fama-French-Carhart four-factor model-adjusted returns. *Tone_{i,t}* represents the tone measure. All regressions use stock-fixed effects and incorporate firm-level and macro-level controls. Firm-level controls include illiquidity and volatility measures, weekly cumulative stock returns up to day $t-1$ before the

(a) Stock-price responses around committee meetings

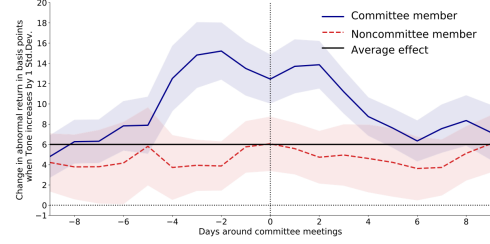


Figure 4
Daily asset price responses around committee meeting days
This figure shows the daily estimated effect of politicians' tweets about a company during the period spanning 9 days before and 9 days after the committee meeting day. Day 0 denotes the actual committee meeting day. The blue line represents the estimated effect of politicians' tweets specifically from the corresponding committee, while the red line represents the estimated effect of politicians' tweets not affiliated with the committee, covering the same time frame. We use a daily event window to compute the stock price responses. The blue and red shadings indicate the 95% error bands.

(b) Daily responses around meeting days

Read:

- Table 4 shows that the benchmark abnormal-return tone coefficient is 2.20 bps.
- On committee meeting days, the high-frequency effect increases: the $\text{Tone} \times \text{Meeting}$ interaction is 0.76 bps with a t -statistic of 2.65.
- When the tweeting politician is a member of the relevant committee, the interaction is 1.38 bps with a t -statistic of 2.49.
- When committee membership and text similarity are both used, the triple interaction remains positive and significant.
- In daily FF4-adjusted returns, committee-member and text-similar tweets add around 4.50 bps to the baseline tone effect.
- Figure 4 shows that committee-member tweets are especially informative in the days surrounding committee meetings.

Economic message. The meeting-day results connect the tweet response to policy news. Tweets by politicians closest to the legislative process carry more price-relevant information.

5.5 Table 5: Bills, legislative effectiveness, and industry spillovers

Table 5
News about bills around committee meeting days

Dependent variable	Daily FF4 adj. returns			Daily industry abnormal returns
Regressors	(1)	(2)	(3)	(4)
$Tone_{i,t}$	6.05 [2.54]	6.05 [2.54]	6.05 [2.54]	1.58 [1.67]
$Tone_{i,t} \times Meeting_t$	2.72 [2.64]	1.12 [1.86]	1.12 [1.86]	1.77 [1.73]
$Tone_{i,t} \times Meeting_t$, and $Bill_t$		2.43 [2.01]	2.33 [1.87]	1.89 [2.27]
$Tone_{i,t} \times Meeting_t$, $Bill_t$ and $Les_{i,t}$			1.75 [1.91]	0.96 [1.97]
1	-6.76 [-1.52]	-6.76 [-1.52]	-6.76 [-1.52]	-7.10 [-1.79]
$Meeting_t$	0.71 [0.13]	0.77 [0.13]	0.77 [0.13]	0.87 [0.18]
$Meeting_t$ and $Bill_t$		5.41 [0.37]	4.21 [0.13]	6.97 [1.89]
$Meeting_t$, $Bill_t$, and $Les_{i,t}$			5.89 [0.97]	-1.14 [-0.07]
R -squared (%)	0.78	0.79	0.82	0.54
Observations	11,355	11,355	11,355	11,321
Firm-level controls	Yes	Yes	Yes	Yes
Macro-level controls	Yes	Yes	Yes	Yes

Moving through columns 1 to 3, we progressively introduce more refined regressors. In column 1, $Meeting_t$ is a dummy variable that equals one if a tweet coincides with a committee meeting and zero otherwise. In column 2, $Meeting$ and $Bill_t$ is a dummy variable that takes the value of one when a committee meeting discusses a specific bill and zero otherwise. In column 3, $Meeting$, $Bill$, and $Les_{i,t}$ take the value of 1 on committee meeting days discussing a bill, and if the LES value for politician i is above the median LES score (calculated using the scores of all members of Congress in the same chamber and corresponding Congress). The dependent variable in columns 1 to 3 is the Fama-French-Carhart four-factor model-adjusted returns. In column 4, the dependent is the industry-adjusted returns obtained by subtracting the industry return from the aggregate market return. $Tone_{i,t}$ represents the tone measure. All regressions use stock-fixed effects and incorporate firm-level and macro-level controls. Stock returns are in basis points, and $Tone_{i,t}$ is standardized. Standard errors are clustered at the firm and day levels, with t -statistics in brackets. R -squared statistics are reported as percentages.

Figure 5: News about bills around committee meeting days

Read:

- The baseline daily FF4 tone coefficient is 6.05 bps.
- The Tone $\times$ Meeting interaction is 2.72 bps with a t -statistic of 2.64 in column 1.
- When the meeting discusses a specific bill, the added interaction is 2.43 bps with a t -statistic of 2.01.
- When the politician has above-median Legislative Effectiveness Score, the bill-related interaction remains positive.
- Column 4 shows industry-level spillovers: Tone $\times$ Meeting $\times$ Bill is 1.89 bps with a t -statistic of 2.27.

Economic message. Tweets tied to specific bills convey policy information that affects both the targeted firm and related firms in the same industry.

6 Short summary

- The paper studies whether individual politicians' social media posts affect asset prices.
- The authors collect congressional Twitter data from 2013 to 2020.
- They identify 11,602 tweets that express opinions about 454 Russell 3000 firms.
- Tweet tone is measured using positive and negative financial dictionaries and then standardized.
- The benchmark high-frequency event window is $[-1, +5]$ minutes around the tweet.
- A one-standard-deviation higher tone raises stock prices by about 2 bps within minutes.

- Effects grow to about 7–8 bps at the daily horizon.
- A long-short portfolio based on supportive versus critical tweets earns large monthly abnormal returns.
- The strongest effects come from tweets connected to committee meetings, committee members, text-similar meeting topics, and bills.
- Industry-level responses show that bill-related tweets reveal information relevant beyond the single mentioned firm.
- The main conclusion is that congressional social media is a channel through which political information enters prices.

7 Conclusion

7.1 Core conclusion

Bianchi, Gómez-Cram, and Kung show that individual members of Congress move asset prices when they tweet viewpoints about specific firms. The effect is visible within minutes, persists over longer horizons, and is strongest when tweets are connected to the legislative process.

7.2 Economic intuition

Markets care about future policy. Congressional tweets can reveal information about politicians' positions, committee discussions, bills, and future legislative actions. A supportive tweet may signal favorable policy prospects; a critical tweet may signal regulatory, legislative, or reputational risk. Because the tweets are public and timestamped, their information content is quickly incorporated into prices.

7.3 Takeaway

The paper turns congressional Twitter accounts into a high-frequency political-news laboratory. Its central message is that political communication affects asset prices not only through formal votes or enacted legislation, but also through the regular flow of public viewpoints from individual politicians.